ML PRACTICAL 2

Email Spam Classification using K-Nearest Neighbors (KNN) and Support Vector Machine (SVM)

Aim

To classify emails as spam or not spam using KNN and SVM algorithms, and compare their performance.

Theory

Email spam classification is a machine learning task used to identify unwanted emails. In this practical, the **emails.csv** dataset is used to classify emails into two categories: Spam (1) and Not Spam (0). The **Email No.** column is removed because it is an identifier, while **Prediction** is used as the target variable. The remaining columns are used as input features.

The dataset is divided into 82% training data and 18% testing data. **StandardScaler** is applied to standardize the features before training the models.

Algorithms Used

1. K-Nearest Neighbors (KNN): KNN is a supervised learning algorithm that classifies a data point based on the majority class of its nearest neighbors. In this practical, the model uses 5 nearest neighbors.

2. Support Vector Machine (SVM): SVM is a supervised learning algorithm that finds an optimal decision boundary (hyperplane) to separate different classes. A linear kernel is used in this practical.

Model Evaluation

Both models are trained on the training dataset and tested on unseen data. Their performance is evaluated using **accuracy** and **confusion matrices**. Accuracy measures the proportion of correct predictions, while a confusion matrix shows correct and incorrect classifications for each class.

Result

KNN and SVM algorithms are implemented for email spam classification. Their performance is compared using accuracy scores and confusion matrices.

Conclusion

This practical demonstrates the use of KNN and SVM for email spam detection. Both algorithms classify emails based on their features and allow their classification performance to be compared.

Code note: In the uploaded notebook, the KNN and SVM accuracy variables are assigned to the opposite predictions. Use the corrected assignments below so the printed labels match the models:

```
knn_acc = accuracy_score(y_test, y_knn_pred)

svc_acc = accuracy_score(y_test, y_svc_pred)
```